Extended reading — Week 5: Function Approximation and DQN

STUDY-RL

May 21, 2026

Reading list

1. Mnih et al. 2015 — "Human-level control through deep reinforcement learning"

Citation. Volodymyr Mnih, Koray Kavukcuoglu, David Silver, et al., *Nature* 518:529–533, 2015. arXiv:1312.5602 (2013) for the NeurIPS workshop precursor.

Summary. The paper that started Deep RL. Combines Q-learning with a deep convolutional network, experience replay, a periodically updated target network, and reward clipping. Achieves human-level performance on 29 of 49 Atari games using identical hyperparameters across all games — a level of generality that had not been seen before in RL. The supplementary material contains the precise hyperparameters (RMSProp, 32 batch, 10^6 replay, 10k target update, $\gamma = 0.99$, $\varepsilon : 1 \to 0.1$ over 1M frames).

Why it matters for this week. The reference implementation we copy. Our mini-project differs only in the use of Polyak target updates and Adam over RMSProp — both modern improvements.

> *"We addressed [instabilities in deep Q-learning] with a novel variant of Q-learning, which uses two key ideas. First, we used a biologically inspired mechanism termed experience replay [. . .]. Second, we used an iterative update that adjusts the action values (Q) towards target values that are only periodically updated, thereby reducing correlations with the target."*

Commentary. The paper presents replay and target as "two ideas" but they are two responses to one problem: the violation of i.i.d. Markov-chain Monte Carlo assumptions that the function-approximation convergence theory presumed. Read with the deadly-triad lens, the entire 2013–2015 DQN line is a series of patches to make off-policy deep RL not diverge in practice.

2. Sutton & Barto 2018 — Chapter 11, "Off-policy Methods with Approximation"

Citation. Richard S. Sutton and Andrew G. Barto, MIT Press, 2nd ed. 2018, Chapter 11.

Summary. The cleanest available statement of the deadly triad, with worked examples (Baird's counterexample, the θ-to-2θ example). Introduces gradient-TD (GTD, GTD2, TDC) as principled fixes that avoid the triad by minimising the projected Bellman error via a true gradient. Concludes with a sober assessment: deep RL has chosen the engineering-fix path (replay + targets) over the principled-fix path (gradient-TD); both work, neither is a guarantee.

Why it matters for this week. The right frame for the entire week. Memorise §11.3 (the triad).

> *"The danger of the deadly triad has been recognized since the early days of TD learning with linear function approximation; despite considerable effort it remains a serious issue. The risk is divergence, that is, weights diverge to infinity, and value estimates become unbounded."*

Commentary. This grounds the lecture in classical RL theory rather than letting DQN appear ex nihilo as a deep-learning invention. The triad is also the right generalisation: every week from 5 to 11 is, in some sense, a different patch on the same underlying disease.

3. Baird 1995 — "Residual Algorithms"

Citation. Leemon Baird, *ICML* 1995.

Summary. Constructs the seven-state counterexample now known as Baird's MDP and shows that semi-gradient off-policy linear TD diverges on it, while a true *residual gradient* (full gradient of the squared Bellman error) is stable but biased. Proposes a residual-algorithm family that interpolates the two.

Why it matters for this week. The first principled identification of the deadly triad and the source of the canonical divergence demo (which our Exercise 7 reproduces).

> *"The phenomena described here are not pathological — they will arise in many practical problems — but the algorithms presented in this paper are guaranteed to converge for any MRP with bounded rewards, providing a sound foundation for future work."*

Commentary. A historical note: deep RL essentially ignored Baird's principled fix and instead engineered around the problem. The Schwarzer 2023 BBF paper revives something close to Baird's intuition (resets break the self-reinforcing divergence loop).

4. van Hasselt et al. 2018 — "Deep Reinforcement Learning and the Deadly Triad"

Citation. Hado van Hasselt, Yotam Doron, Florian Strub, Matteo Hessel, Nicolas Sonnerat, Joseph Modayil, arXiv:1812.02648 (2018).

Summary. A careful experimental study of when the triad actually fires in deep RL. Finds that (i) bigger networks help, (ii) target networks are essential, (iii) prioritized replay can *exacerbate* divergence on hard cases, (iv) double-Q helps. Argues divergence is the exception not the rule under modern deep RL practice.

Why it matters for this week. Calibrates intuition: the triad is a real danger, but mature DQN engineering makes it rare in benign tasks. The paper is also a nice study of how to design RL experiments that isolate one variable at a time.

> *"The deadly triad of function approximation, bootstrapping, and off-policy learning was identified as the source of instabilities in early reinforcement learning research. We investigate the prevalence of these instabilities in deep reinforcement learning algorithms by training a Q-learning agent, with several variations [...]. We find that the triad is not always deadly, and that certain components — in particular, target networks — are crucial."*

Commentary. The experimental design here is worth studying as a methodology: when you have an unstable algorithm with five knobs, the only way to learn anything is to sweep them one at a time, large grid, many seeds. Helpful preview for Week 7's PPO "37 implementation details" paper, which is essentially the same exercise for on-policy PG.

5. Farebrother, Machado & Bowling 2018 — "Generalization and Regularization in DQN"

Citation. Jesse Farebrother, Marlos C. Machado, Michael Bowling, NeurIPS Deep RL workshop 2018, arXiv:1810.00123.

Summary. Shows that vanilla DQN overfits its training trajectory — evaluate on shifted Atari mode/start states and performance collapses. Standard regularisation (dropout, weight decay, ℓ_2) recovers most of the gap. Often cited as the start of the "RL needs DL hygiene" line of work that culminated in BBF and SimBa.

Why it matters for this week. The bridge from vanilla DQN (Week 5) to modern-arch DQN (BBF, SimBa, Week 6 and Week 10). Plasticity loss, dormant neurons, and capacity collapse all trace to this paper.

> *"The lack of generalization in deep RL is real and severe: standard agents trained on hundreds of millions of frames on a single game can fail catastrophically when tested in slightly different versions of the same game."*

Commentary. A useful sobering counterweight to Mnih 2015's optimism. The right mental model: vanilla DQN is a tightly-overfit lookup table with neural smoothing; production-grade deep RL agents earn their generalisation through explicit regularisation (BBF resets, SimBa LayerNorm + residuals, distributional value heads).

Sourced from: `research/deeprl_core_sota.md` §1.1–1.2 (DQN family canon), and Sutton & Barto Chapter 11 for the deadly-triad framing.